The 140−τ scalar radix: evidence report

Date: 2026-10-19
Scope: the 17-position nearest-norm 140−τ scalar recoding on the secp256k1 GLV coefficient lattice, its orbit table, and archived Tier 1 paired batch panels. This is a fixed-base scalar-table throughput report, not an end-to-end ECDLP result.

Question and method

Does π = 140−τ, with norm 19,183, give an exact 17-step recoding and a useful time-memory point between the 18-position rolling format and 16-position 190−τ format?

The screen checks all 19,183 residue congruences, sign/unit covariance and the global minimum on all 9,592 sign-canonical residues. The Lean chain checks the radix identity, selected trajectory/terminal bound, finite digit certificate and runtime tie rule. Independent affine and native replays compare the resulting points against ordinary binary scalar multiplication. The frozen panels use 128 scalars, alternating paired batches, separate table-setup accounting, and retained raw arms and adjacent A/A controls. Commands and file hashes are in the README and run manifests.

Curve and subgroup: secp256k1, `EC1P256Csecp256k1ha8dc9f115a15` in `crypto:docs/curves/registry.json`; $y^2=x^3+7 \bmod p$, prime subgroup order n = `FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364`, cofactor 1, standard generator `G`. No new curve or isogeny is introduced.

Findings

- **Derived and exhaustively screened:** the 140−τ radix has norm 19,183 and the selected rule reaches the reported terminal bound after 17 steps. The digit and terminal orbit tables contain 3,198 and 3,677 seeds. The measured native table retains 4,178,836 bytes, including metadata.
- **Independently replayed:** the affine path checks 388 scalars, and native exhaustive-residue plus 128-case holdout tests matched binary multiplication. Lean proves the integer digit/trajectory properties; the candidate-specific formal point-group bridge remains open.
- **Measured, A/A-qualified hot-table panels:** RunPod `cpu3c`, instance `thyvmdqmzsz5ik` in EU-RO-1 (8 vCPU/16 GB). The captured host record reports x86_64 Ubuntu 24.04.4 under Docker on AMD EPYC 9655 96-Core Processor. The worker was pinned to CPU 13 from `{2,13,40,74,98,109,136,170}`; one NUMA node was reported. CPU quota was unlimited; host exclusivity and memory policy were unverified, and physical memory type/speed/channel data remain unknown. Both 96-pass panels completed 96 arms/48 pairs with stable artifacts and no A/A failures. Ratios are 140−τ/reference hot-table throughput; the repeat-bootstrap 95% intervals describe repeats on this single allocation:

Candidate / reference	Operations per arm	Paired median ratio	95% repeat interval	Retained bytes (reference → candidate)
140−τ / 18-position rolling	12,288	1.0232	[1.0169, 1.0308]	2,448,196 → 4,178,836
140−τ / 16-position 190−τ	12,288	0.9696	[0.9671, 0.9762]	6,829,948 → 4,178,836

In the direct 64-pass 18-versus-17 panel, one adjacent A/A ratio was 1.1362, outside [0.90,1.11]; its descriptive median was 1.0271 but its qualified ratio is null. The longer 96-pass panel was

run under the same allocation and gate and passed. The 140−τ table was faster than the 18-position reference in that qualified hot-table workload, and slower than the 16-position reference while using less retained memory. Setup is charged separately. These measurements do not rank single-call latency or complete scalar multiplication including setup.

Visuals and graph audit

The editable recoding diagram source and SVG show the 17-step path. make_graph.py recreates the paired throughput plot from the two qualified analysis JSON files. The PDF report embeds both figures.

Checked `docs/BENCHMARKS.md`, `docs/perf/PERFORMANCE_INDEX.md`, `docs/perf/weights.json`, and `experiments/volcano-ic/figures/` (`census.svg/.png`, `ecdlp.svg/.png`, `tau.png`). These cover the C/suite kernel index or IC/volcano comparisons; the present Rust fixed-base scalar recoding is outside those records. The experiment-specific throughput graph is new; no existing canonical graph/index entry is affected. The same secp256k1 curve record remains in the sibling crypto registry, with no new route.

Open checks and reproduction

The one-call latency and fresh-allocation replication remain follow-ups; no new benchmark was run for this report. The formal group/point-action bridge is still open, so mathematical point correctness is supported by independent replay, not that theorem. Reproduce the archived screen and analysis with the commands in the README and `analyze_tier1.py`; these consume committed records and do not start cloud jobs.

Source records

- Experiment README, commands, and theorem boundaries
- Qualified 18-versus-17 panel, qualified 16-versus-17 panel
- Failed-gate 64-pass panel
- RunPod raw panels and manifests, 16-versus-17 receipts
- Source revision before this report: PR #905 head `37a57501a3297727abe32764cd86d2ab98af30b3`

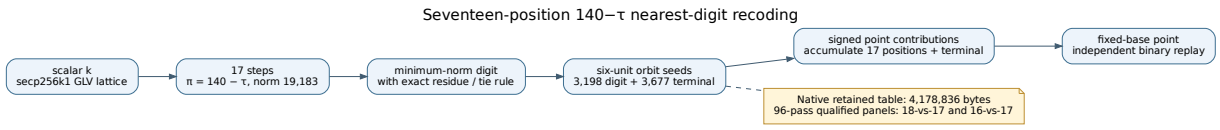


Figure 1: Seventeen-step 140−τ recoding and orbit table

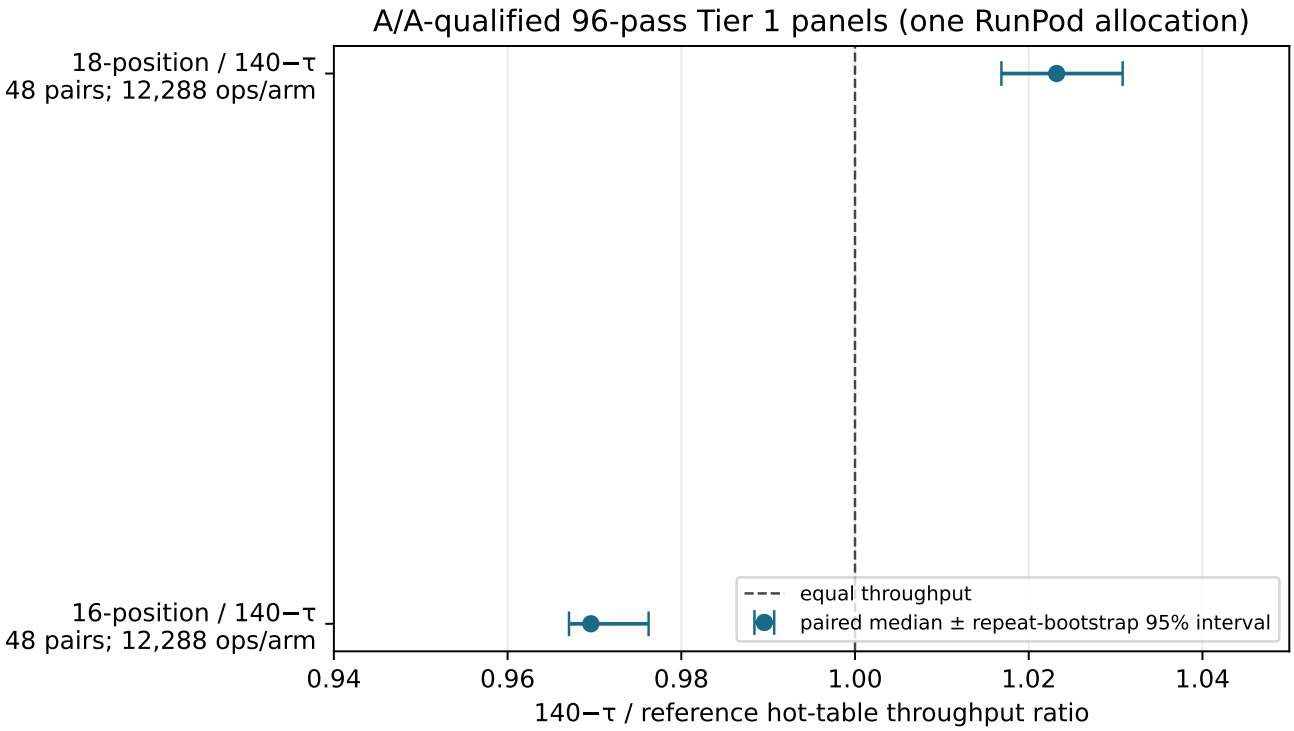


Figure 2: A/A-qualified 140−τ paired throughput panels